

Are β -blockers necessary for patients with heart failure with preserved ejection fraction? PURSUIT-HFpEF registry

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Abstract

Background

The effect of β -blockers on heart failure (HF) with preserved ejection fraction (HFpEF) remains controversial. One proposed reason is the confounding influence of common comorbidities such as atrial fibrillation (AF) and ischaemic heart disease (IHD), which may obscure the influence of β -blockers on HFpEF outcomes.

Methods and results

From the PURSUIT-HFpEF registry (UMIN000021831), patients were divided into two groups: AF/IHD (with AF and/or IHD) and non-AF/IHD (without both). Prognosis was compared between β -blocker users and non-users in each group. Inverse probability of treatment weighting (IPTW) was performed as the primary adjustment method. In AF/IHD cohort ($n = 687$; β -blocker 361, non- β -blocker 326), outcomes did not differ for the composite endpoint, all-cause death, or HF rehospitalization. In contrast, in non-AF/IHD patients ($n = 537$; β -blocker 203, non- β -blocker 334), β -blocker use was linked to poorer outcomes. In IPTW-weighted analyses, β -blocker use was not associated with the composite endpoint, all-cause death, or HF rehospitalization in the AF/IHD group. In contrast, among non-AF/IHD patients, β -blocker use was associated with a higher risk of all-cause death [$P = .046$, hazard ratio (HR) 1.448, 95% confidence interval (CI) 1.007–2.082] and cardiac death ($P = .001$, HR 2.380, 95% CI 1.406–4.027), as well as a higher risk of cardiac composite outcomes ($P = .039$, HR 1.441, 95% CI 1.018–2.039). Formal interaction testing between β -blocker use and AF/IHD status was not statistically significant across endpoints.

Conclusions

In HFpEF patients without AF or IHD, β -blocker use was associated with higher mortality-related risk, indicating that routine β -blocker use in this subgroup should be interpreted with caution.

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without both AF and IHD (non-AF/IHD patients). Because we focused on the effect of β -blocker in this study, we compared the various factors between the patients with β -blockers (β -blocker group) and those without β -blockers (non- β -blocker group) in the AF/IHD patients and non-AF/IHD patients separately. Comparisons were made between the β -blocker group and non- β -blocker group regarding composite endpoints including all-cause death and HF rehospitalization, all-cause death and HF rehospitalization in AF/IHD group and non-AF/IHD group separately.

Data collection

Investigative cardiologists and trained research nurses documented patients' data, including past medical history, HF rehospitalization, comorbidities, therapeutic procedures, and clinical events, from medical records and direct interviews with patients and their family members during their hospital stay and post-discharge. They also collected physical findings, vital signs, echocardiographic data, laboratory data, and medications. Furthermore, the investigative cardiologists and trained research nurses meticulously followed all study patients to prevent loss to follow-up.

Laboratory measurements

Blood samples were collected on admission. Laboratory measurements, including haemoglobin, albumin, estimated glomerular filtration rate (eGFR), C-reactive protein, NT-pro BNP, low-density lipoprotein (LDL) cholesterol and high-density lipoprotein (HDL) cholesterol, triglyceride, and haemoglobin A1c were conducted using standard methods in the clinical laboratory of the participating hospital.

Echocardiographic data

A comprehensive echocardiographic examination was performed by trained physicians at each institution. The left ventricular diastolic diameter (LVDd), left ventricular systolic diameter (LVDs), intraventricular septal thickness at end-diastole (IVSTd), left ventricular posterior wall thickness at end-diastole (LVPWTd), LVEF, left atrial dimension (LAD), E/e' , and tricuspid regurgitant pressure gradient (TRPG) were obtained in accordance with contemporary recommendations. The LVEF was measured by the modified Simpson method. The E/e' was the mean of the septal and lateral E/e' values.

Medications on admission

The medications administered on admission were assessed to verify the prescription of the following: angiotensin-converting enzyme inhibitors (ACEI) or angiotensin II receptor blockers (ARB), β -blockers, calcium antagonist, MRA, SGLT2 inhibitors, diuretics, antiplatelets, anticoagulants, and statins.

Endpoints

Primary endpoint of this study was a composite of all-cause death and HF rehospitalization. Secondary endpoints were all-cause death and HF rehospitalization considered separately. These endpoints were investigated in AF/IHD patients and non-AF/IHD patients separately. In addition, after adjusting the baseline differences, we also evaluated the cardiac composite endpoints including cardiac death and HF rehospitalization and the cardiac death.

Statistical analysis

Continuous variables are expressed as median [interquartile range] and are compared using the Mann–Whitney U test. Categorical variables are expressed as percentages and were compared using either the chi-square test or Fisher exact test. Primary endpoint, a composite of all-cause death and HF rehospitalization, and death rates were compared between β -blocker group and non- β -blocker group using the Kaplan–Meier method in AF/IHD patients and non-AF/IHD patients separately. Gray's test was performed as death was considered a competing risk when comparing HF rehospitalization between β -blocker group and non- β -blocker group in AF/IHD patients and non-AF/IHD patients separately. Cox proportional hazards models were used to evaluate composite endpoint and all-cause death, and Fine–Gray subdistribution hazard models were used to evaluate HF rehospitalization. To reduce the effect of difference of the backgrounds, inverse probability of treatment weighting (IPTW) based on propensity

scores was used as the primary adjustment method. Propensity scores were estimated using logistic regression including clinically relevant covariates selected *a priori*: age, gender, heart rate, systolic blood pressure, history of diabetes mellitus, hypertension, chronic obstructive pulmonary disease, chronic kidney disease and prior HF hospitalization, haemoglobin, LDL and HDL cholesterol, eGFR, NT-pro BNP, LVDd, LAD, E/e' , use of ACEI/ARB, SGLT2 inhibitors, MRA, diuretics, antiplatelets, and anticoagulants. Stabilized IPTW weights were applied, and covariate balance was assessed using standardized mean differences (SMD). In IPTW-weighted analyses, Cox proportional hazards models with robust variance estimators were used to evaluate the composite endpoint and all-cause death. For HF rehospitalization, cause-specific Cox proportional hazards models were applied, as Fine–Gray subdistribution hazard models showed numerical instability under IPTW weighting. In addition, we also evaluated the cardiac composite endpoints including cardiac death and HF rehospitalization and the cardiac death after adjusting the baseline differences.

To formally assess effect modification by AF and/or IHD (AF/IHD), interaction analyses were performed using IPTW-weighted Cox models including β -blocker use, AF/IHD status, and their interaction term (β -blocker $\times$ AF/IHD), with additional adjustment for major prognostic covariates (age, chronic obstructive pulmonary disease, eGFR, and ACE/ARB use). Proportional hazards assumptions were evaluated using Schoenfeld residuals.

As sensitivity analyses, propensity score matching (1:1 nearest-neighbour matching with a caliper of 0.2) was performed separately in AF/IHD and non-AF/IHD patients using the same covariates. Outcomes after matching were analysed using Cox proportional hazards models or Fine–Gray subdistribution hazard models including major prognostic covariates. All statistical analyses were performed using R software (version 4.0.5; R foundation for Statistical Computing, Vienna, Austria). Statistical significance was defined as $P < .05$.

Results

Study patients

The flowchart illustrating this analysis is presented in [Figure 1](#). A total of 1240 patients were enrolled in the PURSUIT-HFpEF registry between June 2016 and February 2022, with clinical follow-up completed through May 2025. The patients with AF were 557 patients and those with IHD were 213 patients. Among 213 patients with IHD, there were 83 patients with myocardial infarction (39%). As a result, there were 687 AF/IHD patients (83 patients had both AF and IHD) and 537 non-AF/IHD patients as shown in [Figure 1](#). We investigated the AF/IHD patients and non-AF/IHD patients separately in this study. In the AF/IHD patients, β -blocker group consisted of 361 patients and non- β -blocker group consisted of 326 patients. In non-AF/IHD patients, β -blocker group consisted of 203 patients and non- β -blocker group consisted of 334 patients.

AF/IHD patients

The baseline patient characteristics of AF/IHD patients were shown in [Table 1](#). In the overall cohort, β -blocker group consisted of 361 patients and non- β -blocker group consisted of 326 patients. In terms of the type of β -blockers, carvedilol was used in 112 patients and bisoprolol in 198 patients, with these two agents accounting for 310 patients (85.9%) of all β -blocker users. Regarding the clinical data, dyslipidaemia, chronic kidney disease and the prevalence of the history of HF admission were significantly higher in the β -blocker group than non- β -blocker group [176 (48.8) vs 115 (34.3), $P < .001$, 171 (47.4%) vs 122 (37.4%), $P = .009$ and 127 (35.2%) vs 73 (22.4%), $P < .001$, respectively], while the other parameters including the classification of Nohria–Stevenson were similar between the two groups. On the laboratory data, HDL cholesterol and eGFR were significantly lower [46 (39, 59) mg/dL vs 51 (40, 61) mg/dL, $P = .034$ and 44.3 (30.0, 56.0) mL/min/1.73 m² vs 48.5 (31.2, 60.2) mL/min/1.73 m², $P = .016$, respectively] and triglyceride and NT-pro BNP were significantly higher [77 (58, 102) vs 67 (50, 88), $P < .001$ and 3651 (2,080, 6650) pg/mL vs 2789 (1,532, 5415) pg/mL, $P = .002$, respectively] in the β -blocker group

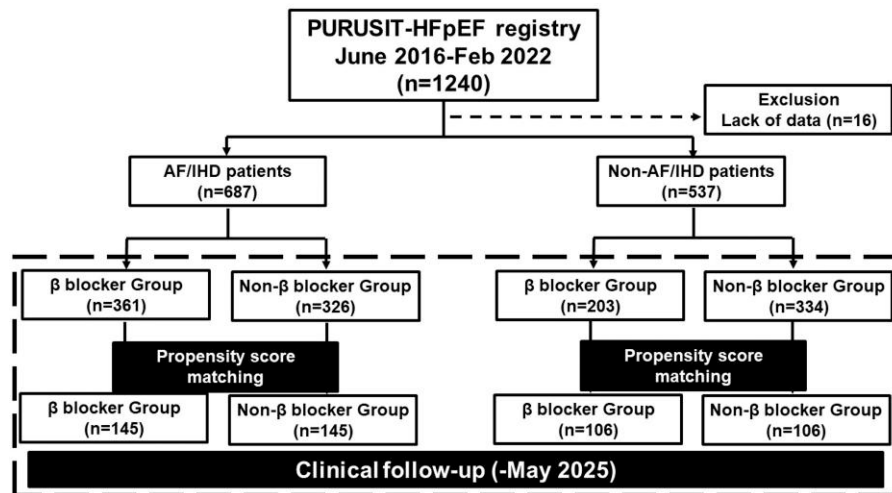


Figure 1 Flow chart of study patients HFpEF, heart failure with preserved ejection fraction; PURSUIT-HFpEF, the Prospective Multicenter Observational Study of Patients With Heart Failure With Preserved Ejection.

than non- β -blocker group. About echocardiographic parameters, there were no significant differences of the parameters. As for the medication, the prevalences of prescription of ACEI/ARB, diuretics, antiplatelets and anticoagulants were significantly higher [204 (56.5%) vs 154 (47.2%), $P = .018$, 239 (66.1%) vs 167 (51.2%), $P < .001$, 143 (39.3%) vs 90 (27.6%), $P < .001$, and 246 (68.0%) vs 165 (50.6%), $P < .001$, respectively] in the β -blocker group than non- β -blocker group. After IPTW adjustment, baseline characteristics were well balanced between the β -blocker and non- β -blocker groups in patients with AF/IHD, with no clinically meaningful differences observed across covariates. All SMDs were ≤ 0.10 or close to this threshold, indicating minimal residual imbalance after weighting (Table 1).

Non-AF/IHD patients

Table 2 revealed that the baseline patient characteristics of non-AF/IHD patients. In the overall cohort, β -blocker group consisted of 201 patients and non- β -blocker group consisted of 334 patients. In terms of the type of β -blockers, carvedilol was prescribed in 89 patients and bisoprolol in 88 patients, accounting for 177 patients (87.2%) receiving either of these two agents. On the clinical data, β -blocker group exhibited a significantly higher prevalences of chronic kidney disease and history of HF admission than non- β -blocker group [102 (50.2%) vs 106 (37.7%), $P < .001$ and 56 (27.6%) and 42 (12.6%), $P < .001$, respectively]. Regarding the laboratory data, β -blocker group had lower values of LDL cholesterol, HDL cholesterol, and eGFR when compared with non- β -blocker group [83 (63, 104) mg/dL vs 91 (72, 109) mg/dL, $P = .004$, 47 (39, 54) mg/dL vs 51 (42, 60) mg/dL, $P = .001$, and 39.1 (24.9, 51.9) mL/min/1.73 m² vs 45.0 (30.4, 19.8) mL/min/1.73 m², $P < .001$, respectively]. As for the echocardiographic parameters, LVDs, LAD and E/e' were significantly higher in β -blocker group than non- β -blocker group [30 (27, 34) mm vs 29 (26, 33) mm, $P = .041$, 44 (40, 49) mm vs 41 (36, 40) mm, $P = .012$, and 17.4 (13.5, 24.0) vs 16.4 (11.6, 31.1), $P < .001$, respectively]. About the medications, the prevalences of prescription of ACEI/ARB, MRA, diuretics, antiplatelets and anticoagulant were significantly higher in β -blocker group than non- β -blocker group [115 (56.7%) vs 141 (42.2%), $P = .001$, 43 (21.2%) vs 43 (12.9%), $P = .015$, 121 (59.6%) vs 136 (40.7%), $P < .001$, 65 (32.0%) vs 69 (20.7%), $P = .004$ and 84 (41.4%) vs 73 (21.9%), $P < .001$, respectively], while the other medications were similar between the two groups. In the non-AF/IHD cohort,

IPTW similarly achieved adequate covariate balance between β -blocker users and non-users. SMDs for all baseline variables were ≤ 0.10 or near this value (Table 2).

Composite endpoint, all-cause death, and HF rehospitalization between β -blocker group and non- β -blocker group in AF/IHD patients

The mean duration of follow-up was 751 [243, 1470] days. During this period, 433 composite endpoints, 303 all-cause deaths (139 cardiovascular deaths), and 272 HF rehospitalizations occurred.

In the overall cohort, Kaplan–Meier analysis indicated that the β -blocker group and the non- β -blocker group had comparable risks for the composite endpoint [$P = 0.470$, hazard ratio (HR) = 1.074, 95% confidence interval (CI) = 0.885–1.303] and all-cause mortality ($P = .343$, HR = 1.117, 95% CI = 0.888–1.406) (Fig. 2A and B). Furthermore, Gray's test indicated no significant differences in HF rehospitalization between the two groups ($P = .730$, HR = 0.940, 95% CI = 0.661–1.336) (Fig. 2C).

After IPTW, there were no significant correlations between β -blocker and outcomes including cardiac composite endpoint and cardiac death (Table 3).

Composite endpoint, all-cause death and HF rehospitalization between β -blocker group and non- β -blocker group in non-AF/IHD patients

Among non-AF/IHD patients, the mean duration of follow-up was 1024 [317, 1511] days. As a result, 309 composite endpoints, 222 all-cause deaths (90 cardiovascular deaths), and 171 HF rehospitalizations occurred during the follow-up period.

In the overall cohort of non-AF/IHD patients, Kaplan–Meier analysis showed that β -blocker group had a significant higher risk for composite endpoint as compared with non- β -blocker group ($P = .029$, HR = 1.288, 95% CI = 1.028–1.818) (Fig. 3A), whereas β -blocker group and non- β -blocker group had similar risk for all-cause death ($P = .089$, HR = 1.263, 95% CI = 0.985–1.653) (Fig. 3B). In addition, Gray's test revealed

Table 1 Baseline patient characteristics of AF/HF patients

	Overall				After IPTW			
	β -Blocker group (n = 361)	Non- β blocker group (n = 326)	P value	SMD	β -Blocker group	Non- β -blocker group	SMD	SMD
Clinical data								
Age, years	83 [78, 87]	83 [78, 87]	.790	0	83 [77, 87]	83 [78, 86]	0.009	
Male (%)	168 (46.5)	172 (52.8)	.109	0.126	52.0	51.8	0.004	
Hypertension (%)	309 (85.6)	261 (80.1)	.067	0.146	83.7	83.7	0.001	
Diabetes mellitus (%)	133 (33.8)	108 (33.1)	.337	0.015	38.4	38.3	0.002	
Dyslipidaemia (%)	176 (48.8)	115 (34.3)	<.001	0.294	46.2	40.1	0.122	
Smoking (%)	257 (71.2)	238 (73.0)	.610	0.040	7.6	7.1	0.020	
Chronic obstructive pulmonary disease (%)	32 (8.9)	24 (7.4)	.399	0.055	8.8	8.8	0.001	
Chronic kidney disease (%)	171 (47.4)	122 (37.4)	.009	0.202	43.8	43.2	0.010	
Dialysis (%)	5 (1.4)	3 (0.9)	.728	0.047				
History of HF admission (%)	127 (35.2)	73 (22.4)	<.001	0.283	29.4	30.6	0.025	
Nohria–Stevenson classification (%)								
Dry-warm	9 (2.5)	14 (4.3)						
Wet-warm	346 (95.8)	307 (94.2)	.647	0.077	94.5	93.7	0.036	
Dry-cold	2 (0.6)	1 (0.3)						
Wet-cold	4 (1.1)	4 (1.2)						
Heart rate	84 [70, 102]	86 [70, 107]	.232	0.078	88 [72, 103]	83 [69, 103]	0.005	
Systolic blood pressure (mmHg)	144 [125, 163]	145 [127, 163]	.674	0.036	144 [125, 164]	142 [125, 165]	0.003	
Laboratory data								
LDL cholesterol, mg/dL	79 [65, 99]	80 [66, 101]	.454	0.039	79 [64, 98]	78 [66, 101]	0.041	
HDL cholesterol, mg/dL	46 [39, 59]	51 [40, 61]	.034	0.329	49 [40, 60]	49 [39, 60]	0.010	
Triglyceride, mg/dL	77 [58, 102]	67 [50, 88]	<.001	0.328	73 [57, 99]	68 [51, 92]	0.054	
Haemoglobin, g/L	11.2 [10.0, 12.5]	11.5 [9.8, 12.8]	.533	0.147	11.3 [10.0, 12.6]	11.5 [9.8, 12.8]	0.013	
eGFR, mL/min/1.73 m ²	44.3 [30.0, 56.0]	48.5 [31.2, 60.2]	.016	0.223	44.6 [30.1, 56.4]	44.7 [28.5, 59.2]	0.006	
Haemoglobin A1c, %	6.1 [5.7, 6.8]	5.9 [5.6, 6.6]	.051	0.257	6.0 [5.6, 6.8]	5.8 [5.5, 6.6]	0.045	
Albumin, g/dL	3.6 [3.3, 3.9]	3.5 [3.2, 3.9]	.548	0.207	3.6 [3.2, 3.9]	3.5 [3.2, 3.8]	0.085	
NT-pro BNP, pg/mL	3651 [2080, 6650]	2789 [1532, 5415]	.002	0.274	3716 [2061, 7208]	2856 [1511, 5725]	0.002	
CRP, mg/dL	0.55 [0.18, 1.88]	0.59 [0.19, 2.42]	.321	0.027	0.60 [0.20, 2.11]	0.57 [0.19, 2.48]	0.086	
Echocardiographic parameters								
LVDd (mm)	46 [41, 49]	46 [42, 50]	.219	0	46 [42, 51]	46 [41, 50]	0.001	
LVDs (mm)	30 [27, 34]	30 [27, 33]	.642	0	31 [27, 34]	30 [27, 33]	0.047	
LVEF (%)	60 [55, 64]	60 [55, 65]	.199	0	59 [55, 64]	60 [55, 65]	0.042	
IVSTd (mm)	10 [9, 11]	10 [8, 11]	.754	0	10 [9, 11]	10 [8, 11]	0.109	
LVPWTd (mm)	10 [9, 11]	10 [9, 11]	.701	0	10 [9, 11]	10 [9, 11]	0.134	
LAD (mm)	46 [41, 52]	46 [41, 52]	.665	0	46 [41, 51]	46 [41, 52]	0.017	
E/e'	16 [12, 20]	16 [12, 20]	.605	0	16 [13, 20]	15 [12, 19]	0.021	

Continued

Table 1 Continued

	Overall		P value	SMD	After IPTW		SMD
	β -Blocker group (n = 361)	Non- β blocker group (n = 326)			β -Blocker group	Non- β blocker group	
TRPG (mmHg)	36 [29, 45]	36 [28, 44]	.569	0	35 [28, 44]	35 [28, 42]	0.085
Medications on admission							
ACEI/ARB (%)	204 (56.5)	154 (47.2)	.018	0.186	55.6	55.1	0.012
Ca antagonist (%)	184 (51.0)	160 (49.1)	.647	0.038	52.2	54.2	0.039
SGLT2 inhibitor (%)	12 (3.3)	7 (2.1)	.330	0.074	3.7	3.9	0.015
MRA (%)	86 (23.8)	78 (23.9)	1.000	0.002	24.6	24.8	0.007
Diuretics (%)	239 (66.1)	167 (51.2)	<.001	0.303	60.6	60.2	0.007
Antiplatelets	143 (39.3)	90 (27.6)	<.001	0.248	36.5	37.1	0.013
Anticoagulants	246 (68.0)	165 (50.6)	<.001	0.354	59.6	58.6	0.020
Statins	110 (30.5)	83 (25.5)	.149	0.111	36.4	31.5	0.103

Continuous data are presented as the median (interquartile range). Categorical variables are presented as numbers (percentage). ACEI, angiotensin-converting enzyme inhibitors; AF, atrial fibrillation; ARB, angiotensin II receptor blockers; CRP, C-reactive protein; eGFR, estimated glomerular filtration rate; HDL, high-density lipoprotein; IVSTd, intraventricular septal thickness at end-diastole; LAD, left atrial dimension; LDL, low-density lipoprotein; MRA, Mineralocorticoid receptor antagonist; LVDD, left ventricular diastolic diameter; LVEF, left ventricular ejection fraction; left ventricular posterior wall thickness at end-diastole; LVDs, left ventricular systolic diameter; NT-proBNP, N-terminal pro brain natriuretic peptide; SMD, standardized mean difference; SGLT2, sodium glucose cotransporter 2; TRPG, tricuspid regurgitant pressure gradient.

β -blocker group had significant higher risk of HF rehospitalization than non- β -blocker group ($P = .008$, HR = 1.494, 95% CI = 1.107–2.017) (Fig 3C). After IPTW, β -blocker use was associated with a significantly higher risk of all-cause death ($P = .046$, HR = 1.448, 95% CI = 1.007–2.082), cardiac composite endpoint ($P = .039$, HR = 1.441, 95% CI = 1.018–2.039), and cardiac death ($P = .001$, HR = 2.380, 95% CI = 1.406–4.027), but β -blocker use was not correlated with the other outcomes (Table 3).

Interaction of β -blocker and AF/IHD for outcomes

In interaction analyses using multivariable Cox proportional hazards models including β -blocker use, AF/IHD status, and their interaction term, no statistically significant interaction between β -blocker use and AF/IHD was observed for any endpoint after IPTW adjustment (Table 4 showing β -blockers, AF/IHD, and β -blockers $\times$ AF/IHD data; Supplementary Tables S1–6 showing all parameters in each analysis). Specifically, the β -blocker $\times$ AF/IHD interaction term was not significant for the composite endpoint (P for interaction = .667), all-cause death ($P = .135$), or HF rehospitalization ($P = .837$). After IPTW adjustment, β -blocker use was associated with a higher risk of all-cause death ($P = .020$, HR = 1.86, 95% CI = 1.065–2.073), whereas AF/IHD status itself was independently associated with HF rehospitalization ($P = .020$, HR = 0.604, 95% CI = 0.396–0.923).

Sensitivity analysis

We performed the propensity score matching as a sensitivity analysis. After the propensity score matching, among AF/IHD patients, no significant differences of all the parameters between β -blocker group ($n = 145$) and non- β -blocker group ($n = 145$) were found (Supplementary Table S7). In the propensity score matching cohort, there were 152 composite endpoints, 99 all-cause deaths, and 100 HF rehospitalizations during the follow-up period. In this cohort, treatment with β -blockers was associated with a significantly reduced risk of all-cause mortality ($P = .025$, HR = .628, 95% CI = 0.419–0.943) compared with non- β -blocker treatment (Supplementary Table S8), but there were no correlation between the β -blocker treatment and composite endpoint and HF rehospitalization. After the propensity score matching, among non-AF/IHD patients, there were no significant differences of all the parameters between β -blocker group ($n = 106$) and non- β -blocker group ($n = 106$) (Supplementary Table S9). In the propensity score matching cohort of non-AF/IHD patients, there were 116 composite endpoints, 84 all-cause deaths, and 55 HF rehospitalizations during the follow-up period. In this cohort, there were no correlation between the β -blocker treatment and composite endpoint, all-cause death and HF rehospitalization (Supplementary Table S10).

Discussion

The study demonstrated the following findings. (i) In the overall cohort, among patients with AF/IHD, there were no significant differences in the composite endpoint, all-cause death, or HF rehospitalization between the β -blocker group and the non- β -blocker group. In contrast, among patients without AF/IHD, the β -blocker group had a negative impact on the composite endpoint and HF rehospitalization, while the incidence of all-cause death was similar between the two groups. (ii) After IPTW, among patients with AF/IHD, the β -blocker also showed no favourable outcome, but, among patients without AF/IHD, β -blocker group showed significant worse all-cause-death, cardiac composite endpoints, and cardiac death compared with non- β -blocker group.

In addition, we performed propensity score matching as a sensitivity analysis. After propensity score matching, among patients with AF/IHD, β -blocker group showed a favourable effect for all-cause death but not

Table 2 Baseline patient characteristics of non-AF/IHD patients

	Overall				After IPTW			
	β -Blocker group (n = 203)	Non- β blocker group (n = 334)	P value	SMD	β -Blocker group	Non- β blocker group	SMD	
Clinical data								
Age, years	82 [77, 87]	82 [75, 88]	.838	0.081	83 [77, 88]	82 [76, 88]	0.007	
Male	80 (39.4)	125 (37.4)	.648	0.041	166.1	166.6	0	
Hypertension	182 (90.0)	280 (83.8)	.052	0.172	89.7	88.0	0.053	
Diabetes mellitus	59 (29.1)	100 (35.9)	.845	0.019	32.0	30.5	0.034	
Dyslipidaemia	92 (45.3)	140 (41.9)	.473	0.069	46.8	49.0	0.044	
Smoking	109 (53.7)	198 (59.2)	.209	0.113	50.5	63.3	0.261	
Chronic obstructive pulmonary disease (%)	11 (8.9)	21 (7.4)	.851	0.037	7.7	6.7	0.041	
Chronic kidney disease (%)	102 (50.2)	106 (37.7)	<.001	0.383	37.7	37.3	0.007	
Dialysis (%)	6 (3.0)	4 (1.2)	.189	0.126	1.9	1.2	0.051	
History of HF admission (%)	56 (27.6)	42 (12.6)	<.001	0.381	16.6	17.5	0.023	
Nohria-Stevenson classification (%)								
Dry-warm	8 (3.9)	19 (5.7)						
Wet-warm	182 (89.7)	302 (90.4)	.060	0.026	87.7	87.3	0.010	
Dry-cold	6 (3.0)	5 (1.5)						
Wet-cold	6 (3.0)	8 (2.4)						
Heart rate	78 [64, 94]	78 [61, 95]	.734	0.296	77 [61, 92]	79 [60, 95]	0.049	
Systolic blood pressure (mmHg)	153 [130, 178]	153 [131, 177]	.640	0.123	155 [133, 182]	155 [134, 177]	0.003	
Laboratory data								
LDL cholesterol, mg/dL	83 [63, 104]	91 [72, 109]	.004	0.506	86 [68, 110]	88 [69, 108]	0.017	
HDL cholesterol, mg/dL	47 [39, 54]	51 [42, 60]	<.001	0.725	48 [39, 61]	50 [42, 61]	0.150	
Triglyceride, mg/dL	85 [63, 114]	80 [59, 111]	.244	0.180	87 [63, 116]	82 [59, 111]	0.055	
Haemoglobin, g/L	11.0 [9.7, 12.3]	11.1 [9.7, 12.0]	.698	0.080	11.0 [9.8, 12.3]	11.0 [9.7, 12.5]	0.024	
eGFR, mL/min/1.73 m ²	39.1 [24.9, 51.9]	45.0 [30.4, 59.8]	<.001	0.168	44.2 [30.3, 59.2]	42.4 [29.2, 58.8]	0.016	
Haemoglobin A1c, %	5.7 [5.5, 6.4]	5.8 [5.5, 6.4]	.526	0.069	6.0 [5.6, 6.6]	5.8 [5.5, 6.4]	0.002	
Albumin, g/dL	3.5 [3.2, 3.8]	3.4 [3.1, 3.7]	.102	0.450	3.5 [3.2, 3.7]	3.4 [3.1, 3.7]	0.102	
NT-proBNP, pg/mL	3330 [1780, 8090]	2960 [1535, 6536]	.109	0.232	2870 [1610, 6894]	2880 [1550, 5863]	0.022	
CRP, mg/dL	0.49 [0.17, 1.78]	0.49 [0.17, 1.62]	.883	0.059	0.54 [0.10, 2.21]	0.48 [0.16, 1.77]	0.050	
Echocardiographic parameters								
LVDd (mm)	47 [44, 51]	46 [42, 51]	.213	0.298	46 [44, 50]	47 [43, 51]	0.002	
LVDs (mm)	30 [27, 34]	29 [26, 33]	.041	0.337	29 [27, 33]	29 [26, 34]	0.052	
LVEF (%)	61 [55, 66]	61 [55, 66]	.629	0	61 [56, 66]	62 [57, 67]	0	
IVSTd (mm)	10 [9, 11]	10 [9, 11]	.055	0	10 [9, 11]	10 [9, 11]	0.129	
LVPWTd (mm)	10 [9, 11]	10 [9, 11]	.081	0	10 [9, 11]	10 [9, 11]	0.128	
LAD (mm)	44 [40, 49]	41 [36, 46]	.012	0.899	42 [40, 46]	42 [38, 47]	0.033	
E/e'	17.4 [13.5, 24.0]	16.4 [11.6, 31.1]	<.001	0.166	17 [13, 21]	17 [12, 22]	0.034	

Continued

Table 2 Continued

	Overall			After IPTW		
	β-Blocker group (n = 203)	Non-β blocker group (n = 334)	P value	β-Blocker group	Non-β blocker group	SMD
TRPG (mmHg)	38 [27, 46]	37 [29, 45]	.092	38 [27, 46]	38 [30, 46]	0.184
Medications on admission						
ACEI/ARB (%)	115 (56.7)	141 (42.2)	.001	54.4	55.0	0.011
Ca antagonist (%)	114 (56.2)	169 (50.6)	.214	60.1	54.7	0.108
SGLT2 inhibitor (%)	6 (3.0)	3 (0.9)	.088	1.3	1.1	0.013
MRA (%)	43 (21.2)	43 (12.9)	.015	14.1	13.3	0.024
Diuretics (%)	121 (59.6)	136 (40.7)	<.001	45.2	46.9	0.033
Antiplatelets	65 (32.0)	69 (20.7)	.004	29.8	27.1	0.060
Anticoagulants	84 (41.4)	73 (21.9)	<.001	20.8	21.8	0.025
Statins	66 (32.0)	88 (26.4)	.562	33.3	32.8	0.011

Continuous data are presented as the median (interquartile range). Categorical variables are presented as numbers (percentage). The abbreviations were same as Table 1.

for the composite endpoint and HF rehospitalization. In contrast among patients with non AF/IHD, there were no significant correlations between β-blocker and outcomes. However, these results of propensity score matching may be weak because approximately 20% of SMDs showed over 0.20, which revealed insufficient balance of adjustment of the backgrounds.

Regarding the interaction analysis, formal interaction testing demonstrated no statistically significant effect modification by AF/IHD status on the association between β-blocker use and clinical outcomes. Therefore, differences observed in subgroup-specific analyses should not be interpreted as definitive evidence of differential treatment effects, but rather as descriptive and hypothesis-generating. Nevertheless, AF and IHD represent established indications for β-blocker therapy, and stratified analyses were performed to provide clinical context for interpreting β-blocker use in HFpEF. Importantly, the absence of a significant interaction suggests that the observed associations are more likely related to underlying indications for β-blocker prescription rather than a disease-specific effect in HFpEF.

Effect of β-blocker on patients with HFpEF

The role of β-blockers in HFpEF remains controversial, due to inconsistent study results. A report from a Swedish registry showed that β-blockers were associated with a lower risk of all-cause death in patients with LVEF ≥40%, (8244 patients matched 2:1 for age and propensity score; 5496 β-blocker users and 2748 non-users) but not with a reduction in the composite endpoint of all-cause death or HF hospitalization.⁵ A meta-analysis comprising 25 RCTs demonstrated β-blockers exhibited a decrease in both all-cause and cardiovascular death when compared with a placebo in patients with LVEF >40%, but this study in assessing the use of β-blockers for treating HFpEF may be limited because only nine studies focused on patients with LVEF >50% and the other study included the patients with LVEF >40% or >45%, and some trials had follow-up durations as short as 12 weeks.⁸ In a recent single-centre retrospective cohort study of 20 206 patients with LVEF ≥50% hospitalized for HF, β-blocker therapy was associated with reduced all-cause death at 30 days, 1 year, and 3 years.⁹ In contrast, a TOPCAT subanalysis of 1761 patients with HFpEF demonstrated that β-blocker use was correlated with a higher number of HF rehospitalization among patients with LVEF ≥50% (1567 patients: 89.0%), but not cardiovascular disease mortality.⁶ No such association was observed in patients with LVEF 45–49%. The SENIORS trial (Randomized Trial to Determine the Effect of Nebivolol on Mortality and Cardiovascular Hospital Admission in Elderly Patients With Heart Failure trial) used an LVEF cutoff >35% to define HFpEF and enrolled 752 participants, approximately half with an LVEF <50%.¹⁰ Although the trial reported benefits of β-blocker in all patients, a recent reanalysis did not find a benefit of β-blocker in patients with LVEF ≥50%, potentially due to limited statistical power.¹¹ The open-label, randomized J-DHF trial (Japanese Diastolic Heart Failure) tested carvedilol in 245 patients with LVEF >40% and did not detect any beneficial or adverse effects even after limiting the analysis to patients with LVEF ≥50%.¹² Similarly, OPTIMIZE-HF registry, which analysed 8873 patients with LVEF ≥50%, reported no prognostic advantage from β-blocker therapy.¹³ Actually, no randomized trial to date has demonstrated outcome benefit of β-blockers in HFpEF. Thus, to reduce the effect of difference of the backgrounds, we performed IPTW based on propensity scores in this study.

Variability in previous findings may, in part, reflect heterogeneity in HFpEF definitions; several studies included patients with LVEF 40–50%, whereas others required LVEF ≥50%. Accordingly, in this study, we define HFpEF as LVEF ≥50%. Another potential explanation is the influence of comorbidities common in HFpEF. We defined AF and IHD as the two major comorbidities in HFpEF in this study. β-Blockers remain essential for managing for rate control in

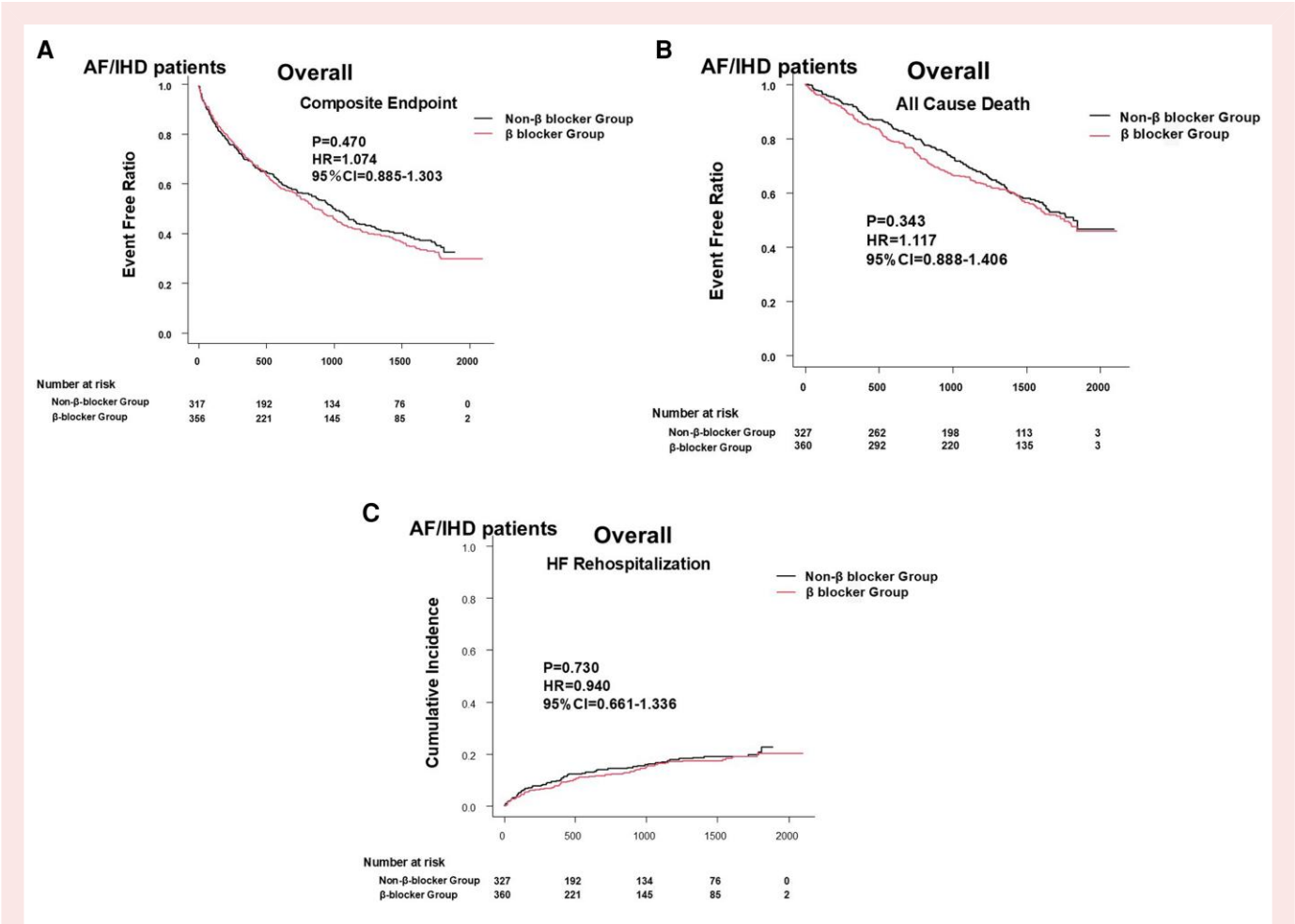


Figure 2 Comparison of composite endpoint, all-cause death and heart failure rehospitalization between β and non- β -blocker groups in AF/IHD patients of overall cohort (a) Composite endpoint, (b) all-cause death, (c) HF rehospitalization. AF, atrial fibrillation; CI, confidence interval; HF, heart failure, HR, hazard ratio; IHD, ischaemic heart disease.

Table 3 Correlation between β -blockers and composite endpoint, death and HF rehospitalization in AF/IHD non-AF/IHD patients after IPTW

Patients	Outcomes	HR	95%CI	P value
AF/IHD	Composite	1.041	0.797–1.358	.769
	All-cause death	1.048	0.763–1.441	.771
	HF rehospitalization	1.010	0.618–1.653	.967
	Cardiac composite	1.289	0.943–1.764	.112
	Cardiac death	1.223	0.797–1.878	.357
Non-AF/IHD	Composite	1.211	0.875–1.675	.249
	All-cause death	1.448	1.007–2.082	.046
	HF rehospitalization	1.035	0.660–1.620	.881
	Cardiac composite	1.441	1.018–2.039	.039
	Cardiac death	2.380	1.406–4.027	.001

AF, atrial fibrillation; HF, heart failure; IHD, ischaemic heart disease; IPTW, inverse probability of treatment weighting.

AF and the treatment of IHD even in HFpEF. Hypertension was also one of the major comorbidities in HFpEF, but there were many antihypertensive drugs including ACEI/ARB and calcium antagonist, and the prevalence of the prescription of β -blocker for hypertension was relatively low in Japan.¹⁴ Accordingly, to consider the pure effect of β -blocker on the comorbidities of HFpEF, we did not select hypertension as the major comorbidities in this study. Consequently, in this study, we defined the two major comorbidities of HFpEF as AF and IHD. Because HFpEF is heterogeneous and typically affects older individuals, the presence of AF and/or IHD may confound the association between β -blocker use and clinical outcomes. To examine the independent effect of β -blockers on HFpEF, we stratified patients into an AF/IHD group (those with AF and/or IHD) and a non-AF/IHD group (those without either condition). The findings suggest that β -blockers may be correlated with worse risk in patients with non-AF/IHD. β -Blockers constitute a key component of evidence-based therapy for HFpEF, independent of ischaemic or non-ischaemic aetiology, as demonstrated in major randomized controlled trials such as CIBIS-II,¹⁵ MERIT-HF,¹⁶ and COPERNICUS,¹⁷ but studies including SENIORS¹⁰ and J-DHF,¹² which enrolled patients with IHD, did not establish a definitive prognostic benefit of β -blockers in HFpEF.

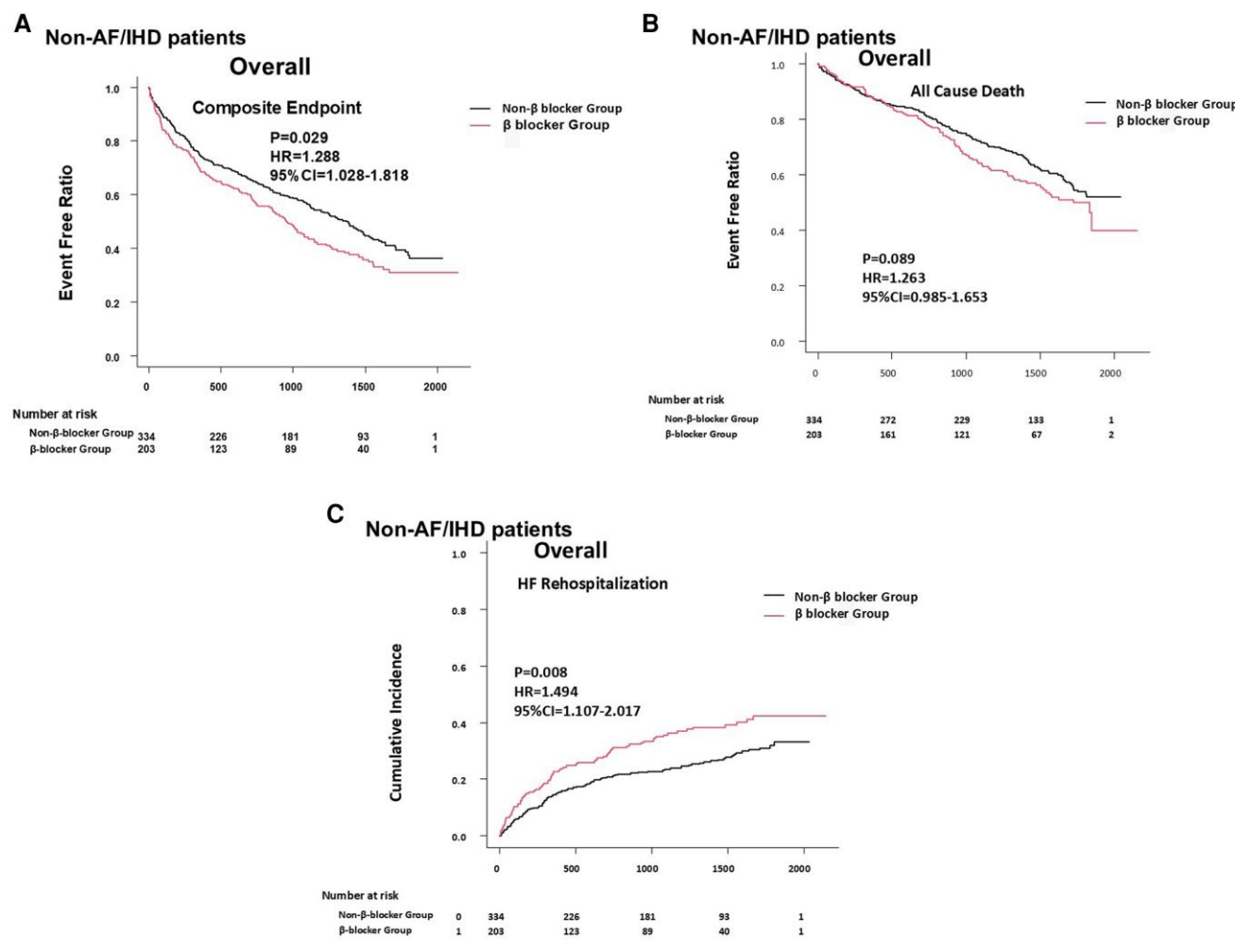


Figure 3 Comparison of composite endpoint, all-cause death and heart failure rehospitalization between β and non- β -blocker groups in non-AF/IHD patients of overall cohort (a) Composite endpoint, (b) all-cause death, (c) HF rehospitalization. The abbreviation was same as Figure 2.

Are β -blockers necessary for all HFpEF patients?

β -Blockers exert chronic effects through pharmacological heart rate reduction. This reduction prolongs diastolic filling, leading to increased ventricular volumes and pressures, thereby elevating ventricular load. The resultant rise in myocardial wall stress may explain the observed elevation of BNP levels in patients receiving β -blockers in population studies.¹⁸ The ELANDD study (Effects of Long-term Administration of Nebivolol on the Clinical Symptoms, Exercise Capacity, and Left Ventricular Function of Patients With Diastolic Dysfunction) which assessed the effect of nebivolol on clinical symptoms and exercise capacity in patients with diastolic dysfunction¹⁹ and the CIBIS-ELD study (Cardiac Insufficiency Bisoprolol Study in Elderly) which evaluated the tolerability of β -blocker uptitration in HFpEF,²⁰ demonstrated β -blocker use was associated with a higher levels of BNP and no benefit of exercise capacity. In the SWEDIC (Swedish Doppler-Echocardiographic) study of patients with HFpEF randomized to receive carvedilol or placebo, BNP levels also increased, and the authors noted an unexpected worsening in HF in the patients treated with carvedilol.⁴ These findings suggest that β -blocker therapy in HFpEF may worsen exercise tolerance, elevate BNP concentrations, and exacerbate HF symptoms. Furthermore, β -blocker can lead to hypotension, bradycardia, and presyncope in the elderly patients, who

represent a substantial proportion of the HFpEF population,⁵ which also may induce the worsening of HF.

On the other hand, it has been reported that in the HF patients with AF and/or IHD, β -blocker is correlated with better prognosis,²¹ but in the patients with HFpEF, especially the elderly patients, the polypharmacy is one of the clinical problems.⁷ Thus, minimizing unnecessary medications is particularly important in this population. In addition, it has been reported that β -blocker cessation for patients with stable HFpEF is safe and leads to marked reductions in NTpro-BNP levels,²² implying that β -blockers may not be necessary for the non-AF/IHD patients with HFpEF.

Moreover, recently, it is controversial whether the β -blocker therapy after myocardial infarction is useful or not. The REBOOT-CNIC trial, which enrolled 8438 post-myocardial infarction patients with LVEF >40%, reported no effect of β -blockers on all-cause mortality, reinfarction, or HF hospitalization.²³ In contrast, the BETAMI-DANBLOCK trial involving 5574 post-myocardial infarction patients with LVEF $\geq$ 40% found that β -blocker therapy reduced the risk of death or major adverse cardiovascular events compared with non- β -blocker therapy.²⁴ In the present study, only 39% of patients with IHD had a history of myocardial infarction, suggesting that β -blocker therapy may be more beneficial in HFpEF patients with IHD unrelated to myocardial infarction, compared with those with prior myocardial infarction. Furthermore, recent published subanalysis of four large HFmrEF/HFpEF trials (I-Preserve,²⁵ TOPCAT,⁶

Table 4 Interaction of β -blocker and AF/IHD for outcomes before and after IPTW

Outcomes	Parameters	Before IPTW			After IPTW		
		HR	95% CI	P value	HR	95% CI	P value
Composite	β -Blockers	0.749	0.554–1.012	.060	1.240	0.926–1.660	.148
	AF/IHD	0.633	0.411–0.975	.038	1.238	0.955–1.605	.106
	β -Blockers $\times$ AF/IHD	1.411	0.956–2.083	.083	0.920	0.629–1.345	.667
All-cause death	β -Blockers	0.778	0.551–1.098	.154	1.486	1.065–2.073	.020
	AF/IHD	0.697	0.420–1.156	.162	1.218	0.895–1.657	.209
	β -Blockers $\times$ AF/IHD	1.326	0.845–2.083	.220	0.714	0.459–1.111	.135
HF rehospitalization	β -Blockers	0.754	0.504–1.128	.170	1.173	0.791–1.741	.427
	AF/IHD	0.278	0.142–0.546	>.001	0.604	0.396–0.923	.020
	β -Blockers $\times$ AF/IHD	1.237	0.672–2.278	.490	1.065	0.586–1.937	.837

The abbreviations were same as Table 1.

PARAGON-HF,²⁶ and DELIVER²) demonstrated that β -blocker therapy was associated with better outcomes among patients with AF, but not among those in sinus rhythm,²⁷ which partially support our results. However, there was an opposite report showing an absence of outcome benefit with β -blocker therapy in patients with AF and HF.²⁸ Currently, β -blockers show no clear benefit in HFpEF regardless of rhythm or IHD, based on available RCTs and meta-analyses. Therefore, further large-scale RCTs are warranted to validate our findings.

Clinical implication

HFpEF is a heterogeneous syndrome in which pharmacological therapy is largely driven by comorbidities rather than disease-specific treatments. In clinical practice, β -blockers are frequently prescribed to patients with HFpEF for established indications such as two major comorbidities: AF or IHD. Based on these considerations, the present IPTW-adjusted analyses suggest that the prognostic associations of β -blocker use in HFpEF differ according to the presence or absence of AF and/or IHD. Importantly, these findings should not be interpreted as evidence supporting a therapeutic benefit of β -blockers in HFpEF itself. Because AF and IHD are easily identifiable in routine clinical practice, this simple stratification may assist clinicians in interpreting observed associations between β -blocker use and outcomes in HFpEF, without implying treatment recommendations.

Limitations

This study has several limitations. First, it did not evaluate the type or dosage of β -blockers administered in detail. However, regarding the type or β -blockers, most β -blockers (approximately 86%) used were either carvedilol or bisoprolol, and prior studies have reported comparable effects between these agents.²⁹ Therefore, the impact of β -blocker type in this study appears minimal. In addition, the OPTIMIZE-HF study reported no association between β -blocker dose and outcomes in HFpEF,³⁰ suggesting that the absence of detailed dosage data may also have had minimal impact on the present findings. Second, this study has inherent limitations related to its observational design, including the potential influence of unmeasured confounders such as indication bias, frailty, disease severity, and medication tolerance. Although propensity score-based methods were applied to reduce baseline imbalances, subgroup analyses, particularly after propensity score matching, were based on relatively small sample sizes and a limited number of events. This substantially reduces statistical power and increases the risk of unstable effect estimates and spurious subgroup findings, including false-negative or false-positive interaction signals. Accordingly, the absence

of statistically significant interactions should be interpreted with caution. In addition, despite careful selection of clinically relevant covariates *a priori*, propensity score models may be susceptible to overfitting when the number of covariates is large relative to the number of outcome events. While IPTW was therefore adopted as the primary analytical approach to improve covariate balance and preserve sample size, residual confounding cannot be fully excluded, and causal inferences should be avoided. Taken together, these limitations highlight that the subgroup-specific findings of this study should be regarded as exploratory and hypothesis-generating rather than definitive evidence of differential treatment effects.

Third, we did not have the data regarding the location of the inclusion. Thus, we did not evaluate whether the location of inclusion was a confounder or not in this study. Fourth, because the multicentre registry in this study was conducted exclusively in Japanese centres, a part of the results may not be applied to Europe and the USA, for examples, the ageing trend in Japan is more remarkable than in Europe and the USA. Japan has the most ageing society worldwide. Therefore, our data may not apply equally to Europe and the USA because our study population is considerably older (mean age 83 years) than those included in major HFpEF trials such as EMPEROR¹ and DELIVER² (mean age approximately 71 years). Thus, age-related differences may affect extrapolation of our findings to Europe or the USA. Fifth, because this study enrolled only patients hospitalized with decompensated HF, the results may not be applicable to stable HFpEF patients managed in the outpatient setting. Finally, the proportion of patients receiving SGLT2 inhibitors was low (2.8% in the AF/IHD group and 1.7% in the non-AF/IHD group, $P = .249$), despite emerging evidence supporting their role as therapeutic agents for HFpEF.^{1,2} This limited use may be attributed to the Japanese national health insurance system only beginning to cover SGLT2 inhibitors for HF in 2021. If SGLT2 inhibitor use had been more common, the interaction between β -blockers and clinical outcomes may have differed, as these agents have been shown to improve prognosis regardless of β -blocker use.^{1,2}

Conclusions

In the PURSUIT-HFpEF registry, our analyses showed that β -blocker use was associated with higher mortality-related risk in patients without AF or IHD, whereas no consistent adverse association was observed in those with these comorbidities. These findings most likely reflect confounding by indication and the absence of established indications for β -blockers in HFpEF without AF or IHD, rather than a disease-specific therapeutic effect. Taken together, our results suggest that routine use

or continuation of β -blockers in HFpEF patients without AF or IHD may warrant careful re-evaluation. Accordingly, these findings should be interpreted cautiously and without inferring a causal relationship.

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Supplementary data

Supplementary data are available at *European Journal of Heart Failure* online.

Declarations

Disclosure of Interest

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Data Availability

The data that support the findings of this study are not openly available due to reasons of sensitivity.

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Ethical Approval

Ethical Approval was not required.

Pre-registered Clinical Trial Number

None supplied.

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